

Dog Tracker Battery Life Analysis

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1 Introduction

This document summarizes the battery life of an ESP32-S3-based dog tracker with a 350 mAh (3.7 V, 1.295 Wh) battery, across three use case sets (high, medium, low consumption) and an averaged daily use. The dynamic firmware adapts to activity levels, resulting in varying battery life based on daily power consumption.

2 Battery Life Results

The dog tracker's battery life is calculated by dividing the 350 mAh capacity by the daily power consumption for each use case set. Below are the results, followed by a summary table.

2.1 Set 1: High Consumption (Active Dog)

Description: A highly active dog (e.g., Border Collie) with frequent walks and movement.

Daily Consumption: 3.725 mAh/day.

Battery Life:

$$\frac{350}{3.725} \approx 93.96 \text{ days.}$$

Components: 4 hr/day BLE (1.20 mAh), 20 hr/day sensor sampling (2.00 mAh), 15 s/day Wi-Fi (0.29 mAh), 20 s/day epilepsy events (0.08 mAh), 2 hr/day light sleep (0.10 mAh), 2 hr/day deep sleep (0.04 mAh), BLE idle (0.01 mAh).

2.2 Set 2: Medium Consumption (Typical Dog)

Description: A moderately active dog (e.g., Labrador) with standard walks and rest periods.

Daily Consumption: 3.06 mAh/day.

Battery Life:

$$\frac{350}{3.06} \approx 114.38 \text{ days.}$$

Components: 2.5 hr/day BLE (0.75 mAh), 17 hr/day sensor sampling (1.70 mAh), 10 s/day Wi-Fi (0.20 mAh), 10 s/day epilepsy events (0.04 mAh), 5 hr/day light sleep (0.25 mAh), 6 hr/day deep sleep (0.12 mAh), BLE idle (0.01 mAh).

2.3 Set 3: Low Consumption (Sedentary Dog)

Description: A low-activity dog (e.g., older Bulldog) with minimal movement and frequent rest.

Daily Consumption: 1.54 mAh/day.

Battery Life:

$$\frac{350}{1.54} \approx 227.27 \text{ days.}$$

Components: 1 hr/day BLE (0.30 mAh), 12 hr/day sensor sampling at 0.5% duty cycle (0.60 mAh), 5 s/day Wi-Fi (0.10 mAh), 5 s/day epilepsy events (0.02 mAh), 7 hr/day light sleep (0.35 mAh), 8 hr/day deep sleep (0.16 mAh), BLE idle (0.01 mAh).

2.4 Averaged Daily Use

Description: A typical dog with a mix of high, medium, and low activity days (e.g., active weekdays, sedentary weekends).

Daily Consumption:

$$\frac{3.725 + 3.06 + 1.54}{3} = 2.775 \text{ mAh/day.}$$

Battery Life:

$$\frac{350}{2.775} \approx 126.13 \text{ days.}$$

Note: Assumes equal weighting (1/3 each) for high, medium, and low use cases.

3 Summary Table

Use Case	Daily Consumption (mAh)	Battery Life (days)
Set 1: High Consumption	3.725	93.96
Set 2: Medium Consumption	3.06	114.38
Set 3: Low Consumption	1.54	227.27
Averaged Daily Use	2.775	126.13

Table 1: Battery life comparison across use cases.

4 Conclusion

The dog tracker's battery life ranges from 93.96 days (high consumption) to 227.27 days (low consumption), with a typical average of 126.13 days. The dynamic firmware ensures adaptability to the dog's activity level, meeting the target of 45+ days across all scenarios. Real-world testing with a power profiler is recommended to validate these estimates.